

Essentially all regression models that we have dealt with thus far have been mean regression models since they relate the predictor η of a regression model to only one specific quantity of the response y , namely the expected value. For example, in case of a generalized linear model (or its extensions) with predictor η , we have

$$E(y) = h(\eta),$$

where h is a known response function. The distribution of the response was then, depending on this mean parameter, completely characterized (sometimes up to a scale parameter common to all observations and potentially with some prespecified weights) by the regression model. In case of Gaussian responses $y \sim N(\mu, \sigma^2)$, we have already discussed in Sects. 2.9 and 4.1.3 that it can be useful to not only model the expectation μ , but also the variance σ^2 in terms of covariates to deal with heteroscedastic errors. We next develop a regression approach that is completely distribution-free to estimate the effect of a regressor on the *quantiles* of the response distribution. As such, we will also relax a number of assumptions of usual normal regression.

As an example to illustrate the utility of such regression models, consider the effects of living area and year of construction on the monthly net rent in the Munich rent index data. Figure 10.1 shows the corresponding scatter plots together with parametric least squares fits (for a linear model in case of living area and a quadratic fit in case of year of construction). Note that we are not dealing with the rent per square meter that we have previously used for most analyses in this book, but rather with the rent itself since this variable is better suited to illustrate the usefulness of quantile regression. Figure 10.1 clearly indicates that a simple mean regression line obtained from least squares is not sufficient to completely describe the dependency between rent and the covariates. In particular, the variance of the response is changing with the covariate values, at least in case of the living area where we find increased variability for larger apartments. This could be taken into account in a location-scale model

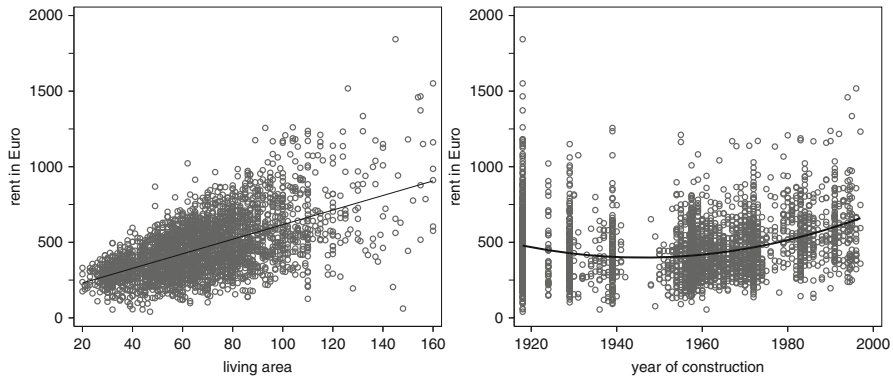


Fig. 10.1 Munich rent index: scatter plots of rents in Euro versus living area (*left panel*) and year of construction (*right panel*) together with a linear (*left panel*) and a quadratic (*right panel*) least squares fit

$$y = \mathbf{x}'\boldsymbol{\beta} + \exp(\mathbf{z}'\boldsymbol{\alpha})\varepsilon, \quad \varepsilon \sim N(0, 1), \quad (10.1)$$

where not only the mean $E(y) = \mathbf{x}'\boldsymbol{\beta}$, but also the variance $\text{Var}(y) = \exp(\mathbf{z}'\boldsymbol{\alpha})^2$ depends on covariates. We have already discussed similar approaches in Sect. 4.1.3 (including estimation procedures) although we have modeled the variance there instead of the standard deviation. In any case, this model still may not be flexible enough, e.g., if in addition to the variance, the skewness of the response also depends on covariates. Instead of explicitly modeling the variance of the response in addition to the mean, our aim in this chapter is to introduce a regression tool to portray essentially all of the characteristics of the response in terms of covariates at once.

We therefore focus on the quantiles of the response distribution and relate these quantiles to covariate effects. The basic idea is that a dense set of quantiles completely describes any given distribution. Hence, when computing enough quantile regression results, we will be able to analyze virtually any property of the response distribution based on these results. Prior to introducing the details corresponding to estimation, we outline some of the advantages of quantile regression:

- Quantile regression allows investigation of covariate effects, not only on the mean of a response variable but on the complete conditional distribution of the response given covariates.
- Quantile regression avoids some of the restrictive assumptions of the linear model (or more generally mean regression models). More specifically, we will not require homoscedasticity or a specific type of distribution for the responses (or equivalently the error terms).
- In applications, there often is a genuine interest in regression quantiles that describe “extreme” observations in terms of covariates. For example, in case of the Munich rent index, one is often interested in interval estimates that cover a range of “usual” rents for a given set of covariates. While such a prediction interval can also be computed from mean regression results (under certain

assumptions on the error variance), it may be more strategic to directly aim at the estimation of the interval boundaries based on regression quantiles.

To illustrate the last point, it is important to note that, in fact, any regression model can also determine estimates for the quantiles of the response. For example, in case of a normally distributed response as in the location-scale model (10.1), the τ -quantile, $\tau \in (0, 1)$, is given by

$$\mathbf{x}'\boldsymbol{\beta} + z_\tau \exp(\mathbf{z}'\boldsymbol{\alpha}), \quad (10.2)$$

where z_τ denotes the τ -quantile of the standard normal distribution. However, the normal distribution for the error terms implies that the quantiles are symmetric around $\mathbf{x}'\boldsymbol{\beta}$, i.e., $\mathbf{x}'\boldsymbol{\beta} + z_\tau \exp(\mathbf{z}'\boldsymbol{\alpha}) = \mathbf{x}'\boldsymbol{\beta} - z_{1-\tau} \exp(\mathbf{z}'\boldsymbol{\alpha})$. If, in addition, the variance is in fact constant and does not depend on covariates such that $\exp(\mathbf{z}'\boldsymbol{\alpha}) = \sigma^2 = \text{const}$, the quantile curves for varying covariates are parallel to each other and only shifted according to the standard normal quantiles. Consequently, more flexible approaches that aim at the direct estimation of quantiles appear to be more appropriate than first starting with a parametric regression model.

10.1 Quantiles

Before actually introducing quantile regression models, we recall the definition of quantiles, both theoretically for distributions and for observed random samples, and discuss some properties that are useful to understand quantile regression. The theoretical quantiles q_τ , $\tau \in (0, 1)$, of a random variable y are commonly and implicitly defined by the equations

$$P(y \leq q_\tau) \geq \tau \quad \text{and} \quad P(y \geq q_\tau) \geq 1 - \tau,$$

i.e., the probability of observing a value below (or equal to) q_τ should be (at least) τ while the probability of observing a value above (or equal to) q_τ should be (at least) $1 - \tau$. For quantile regression, it is useful to reformulate this implicit definition as the optimization problem

$$q_\tau = \arg \min_q E(w_\tau(y, q)|y - q|) \quad (10.3)$$

with weights

$$w_\tau(y, q) = \begin{cases} 1 - \tau & y < q, \\ 0 & y = q, \\ \tau & y > q. \end{cases}$$

Such weights define q_τ as the minimizer of an asymmetrically weighted absolute deviations criterion; see the appendix for a proof of the equivalence. Note that the weights $w_\tau(y, q)$ are defined differently for values above and below q and thus effectively shift the solution upwards or downwards depending on the choice of τ .

If y is continuous with strictly increasing cumulative distribution function $F(y)$ and density $f(y)$, the theoretical quantile is unique and is simply given by the inverse of the cumulative distribution function evaluated at τ , i.e.,

$$q_\tau = F^{-1}(\tau) \quad \text{and} \quad F(q_\tau) = \tau.$$

The function $Q(\tau) = F^{-1}(\tau) = q_\tau$ is also called the quantile function of the distribution of y . Note that the assumption of a strictly increasing cumulative distribution function is not very restrictive since it basically implies that the distribution is continuous and that there are no gaps in the domain of the response where the density is zero.

Empirical quantiles correspond to the estimated quantiles $\hat{q}_\tau$ determined from an i.i.d. sample $y_1, \dots, y_n$ of observations from the corresponding distribution. An implicit definition can again be given as follows: At least a fraction of τ observations should be smaller or equal than $\hat{q}_\tau$ and at least a fraction of $1 - \tau$ observations should be larger or equal than $\hat{q}_\tau$, i.e.,

$$\frac{1}{n} \sum_{i=1}^n I(y_i \leq \hat{q}_\tau) \geq \tau \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n I(y_i \geq \hat{q}_\tau) \geq 1 - \tau,$$

where $I(\cdot)$ denotes the indicator function. For estimation purposes and in particular the generalization to the regression context, it is again better to use the equivalent definition as the solution of an optimization criterion, where

$$\hat{q}_\tau = \arg \min_q \sum_{i=1}^n w_\tau(y_i, q) |y_i - q| \quad (10.4)$$

is the empirical analogue to Eq. (10.3).

An interesting quantity that is associated with the estimation of quantiles is the *influence function* $I(y)$, given by the weights associated with a specific value of y in the estimating equations for determining a quantile:

$$I(y) = \begin{cases} 1 - \tau & y < q_\tau, \\ 0 & y = q_\tau, \\ \tau & y > q_\tau. \end{cases}$$

The influence function is studied in detail in the area of robust statistics, since it measures the impact of an observation y on the determination of a quantity of interest, the τ -quantile in our case. The influence function related to the quantile function can be interpreted as follows: The distance between y and q_τ does not affect the influence function, which only depends on the sign of the deviation between y and q_τ . This gives a theoretical justification for the robustness of quantiles with respect to outliers since, regardless of the size of an observation, the impact is

limited to $\max(\tau, 1 - \tau)$. For comparison, the influence function for the arithmetic mean is given by

$$I(y) = |y - \mu|,$$

where $\mu = E(y)$ and therefore the impact of values further away from the expectation is much larger than the impact of values that are close to μ .

10.2 Linear Quantile Regression

10.2.1 Classical Quantile Regression

Estimation

We will now transfer regression techniques from mean regression to regression for quantiles. Recall that the standard linear regression model takes the form

$$y = \mathbf{x}'\boldsymbol{\beta} + \varepsilon$$

and (in addition to further properties of the error terms) assumes that $E(\varepsilon) = 0$. This implies that the regression coefficients impact the expectation of the response since

$$E(y) = \mathbf{x}'\boldsymbol{\beta} + E(\varepsilon) = \mathbf{x}'\boldsymbol{\beta}.$$

Hence, the assumption on the expectation of the error term implies a specific interpretation of the regression coefficients. For quantile regression, we also assume the model

$$y = \mathbf{x}'\boldsymbol{\beta}_\tau + \varepsilon_\tau$$

but will now make specific assumptions on the error term for the quantile τ of interest to determine quantile-specific regression coefficients $\boldsymbol{\beta}_\tau$ and therefore also explicitly add a τ subscript to both quantities. If we assume that the cumulative distribution function F_{ε_τ} of the error term ε_τ fulfills $F_{\varepsilon_\tau}(0) = \tau$ or equivalently that the τ -quantile of the error term is zero, then this implies

$$\tau = F_{\varepsilon_\tau}(0) = P(\varepsilon_\tau \leq 0) = P(\mathbf{x}'\boldsymbol{\beta}_\tau + \varepsilon_\tau \leq \mathbf{x}'\boldsymbol{\beta}_\tau) = P(y \leq \mathbf{x}'\boldsymbol{\beta}_\tau) = F_y(\mathbf{x}'\boldsymbol{\beta}_\tau)$$

and therefore the τ -quantile of the response is given by the predictor $\mathbf{x}'\boldsymbol{\beta}_\tau$. Estimation of the quantile-specific regression coefficients $\boldsymbol{\beta}_\tau$ is then achieved by generalizing the asymmetrically weighted absolute error criterion (10.4) to

$$\hat{\boldsymbol{\beta}}_\tau = \arg \min_{\boldsymbol{\beta}} \sum_{i=1}^n w_\tau(y_i, \eta_{i\tau}) |y_i - \eta_{i\tau}|, \quad (10.5)$$

10.1 Quantile Regression

Model

$$y_i = \mathbf{x}_i' \boldsymbol{\beta}_\tau + \varepsilon_{i\tau}, \quad i = 1, \dots, n,$$

with assumptions

1. $F_{\varepsilon_{i\tau}}(0) = \tau$.
2. $\varepsilon_{1\tau}, \dots, \varepsilon_{n\tau}$ are independent.

Estimation of Regression Coefficients

The regression coefficients $\boldsymbol{\beta}_\tau$ are determined by minimizing

$$\sum_{i=1}^n w_\tau(y_i, \eta_{i\tau}) |y_i - \eta_{i\tau}|,$$

where $\eta_{i\tau} = \mathbf{x}_i' \boldsymbol{\beta}_\tau$.

Software

- R package `quantreg`: Implements linear programming for determining $\hat{\boldsymbol{\beta}}_\tau$. See [Koenker \(2005\)](#) for details.
- R package `mboost`: Implements functional gradient descent boosting for determining $\hat{\boldsymbol{\beta}}_\tau$. See [Fenske, Kneib, and Hothorn \(2011\)](#) for details.

where $\eta_{i\tau} = \mathbf{x}_i' \boldsymbol{\beta}_\tau$ and

$$w_\tau(y_i, \eta_{i\tau}) = \begin{cases} 1 - \tau & y_i < \eta_{i\tau}, \\ 0 & y_i = \eta_{i\tau}, \\ \tau & y_i > \eta_{i\tau}. \end{cases}$$

Basically, instead of assuming that the average error in the regression model is zero, we assume that the weighted “average” of the errors with weights defined as above is zero. A summary of the quantile regression setup is given in Box 10.1.

The most important feature of quantile regression to be noted here is its generality with respect to the error distribution. Apart from independence of the error terms between individual observations and the quantile restriction, there are no further assumptions on the corresponding distribution. In particular, the error terms may even follow different types of distributions and further, are not assumed to be homoscedastic. While usual least squares estimation in mean regression can also be accomplished without assuming specific types of distributions for the error terms, it always requires constant variances across the errors (unless generalized

to weighted least squares with weights specified in advance). Such generality of quantile regression is one of its specific strengths and essentially leads us to a distribution-free regression approach.

To minimize the asymmetrically weighted absolute error criterion (10.5), we would like to proceed as in the case of mean regression, i.e., taking partial derivatives with respect to the regression coefficients and setting the derivatives to zero. Unfortunately, this is no longer possible since Eq. (10.5) is not differentiable in the origin. We therefore expand the estimation problem by introducing $2n$ auxiliary variables

$$\begin{aligned} u_{i\tau} &= (y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)_+, \\ v_{i\tau} &= (\mathbf{x}'_i \boldsymbol{\beta}_\tau - y_i)_+, \end{aligned}$$

where $(x)_+ = \min(x, 0)$ and therefore $\varepsilon_{i\tau} = u_{i\tau} - v_{i\tau}$. The minimization problem (10.5) can then be rewritten as

$$\sum_{i=1}^n w_\tau(y_i, \eta_{i\tau}) |y_i - \eta_{i\tau}| = \sum_{i=1}^n \tau u_{i\tau} + \sum_{i=1}^n (1 - \tau) v_{i\tau} = \tau \mathbf{1}' \mathbf{u}_\tau + (1 - \tau) \mathbf{1}' \mathbf{v}_\tau,$$

given the constraint that

$$y_i = \eta_{i\tau} + \varepsilon_{i\tau} = \mathbf{x}'_i \boldsymbol{\beta}_\tau + u_{i\tau} - v_{i\tau},$$

or in matrix notation

$$\mathbf{y} = \mathbf{X} \boldsymbol{\beta}_\tau + \mathbf{u}_\tau + \mathbf{v}_\tau,$$

with $\mathbf{u}_\tau = (u_{1\tau}, \dots, u_{n\tau})'$, $\mathbf{v}_\tau = (v_{1\tau}, \dots, v_{n\tau})'$, and $\mathbf{1} = (1, \dots, 1)'$. In summary, the optimization problem after introducing the auxiliary variables is now given by

$$\min_{\boldsymbol{\beta}_\tau, \mathbf{u}_\tau, \mathbf{v}_\tau} \{ \tau \mathbf{1}' \mathbf{u}_\tau + (1 - \tau) \mathbf{1}' \mathbf{v}_\tau \mid \mathbf{X} \boldsymbol{\beta}_\tau + \mathbf{u}_\tau - \mathbf{v}_\tau = \mathbf{y} \}.$$

This is a constrained minimization problem with polyhedral constraints, i.e., the constraint defines a geometric object with flat faces and straight edges, such that the constrained problem corresponds to the original quantile restriction, but written in terms of the auxiliary variables. After augmenting the auxiliary variables and given the constraints, the minimization problem is now linear in the parameters and can therefore be tackled with linear programming techniques that allow to incorporate the polyhedral constraints. Refer to Lange (2000) for a brief introduction to linear programming with a focus on statistics-related applications and Lange (2004) for a detailed exposition.

An alternative estimation approach for minimizing the asymmetrically weighted absolute deviations criterion (10.5), which utilizes the boosting approach developed in Sect. 4.3, has been proposed in Fenske et al. (2011). This approach only requires the following modification: substitute the residuals that have served as working

responses in (linear model) boosting with the corresponding negative gradients of the loss function

$$\rho(y, \eta) = w_\tau(y, \eta)|y - \eta|,$$

which define the individual contributions to Eq. (10.5). These specific negative gradients are given by

$$u = \frac{\partial}{\partial \eta} \rho(y, \eta) = \begin{cases} \tau - 1 & y < \eta, \\ 0 & y = \eta, \\ \tau & y > \eta. \end{cases} \quad (10.6)$$

The case $y = \eta$ occurs with probability zero, and therefore the non-differentiability of Eq. (10.5) does not cause a problem. Apart from that minor change, the rest of the boosting approach, and also the extensions discussed in Sect. 9.7, can immediately be used.



It is interesting to compare quantile regression estimation and least squares estimation in somewhat more detail. For the latter, we are considering the optimization problem

$$\min_{\beta} (y - X\beta)'(y - X\beta)$$

and the solution $\hat{\beta}$ is characterized by the normal equations

$$X'(y - X\beta) \stackrel{!}{=} \mathbf{0}.$$

For quantile regression, the derivative of the optimization criterion

$$R(\beta_\tau) = \sum_{i=1}^n w_\tau(y_i, \eta_{i\tau})|y_i - \eta_{i\tau}|$$

only exists at points with $y_i - \eta_{i\tau} \neq 0$, $i = 1 \dots, n$, since the absolute value function is differentiable everywhere except in the origin. However, we can always define direction-specific derivatives in direction $\mathbf{w} \in \mathbb{R}^p$

$$\left. \frac{\partial}{\partial t} R(\beta_\tau + t\mathbf{w}) \right|_{t=0} = - \sum_{i=1}^n \psi_\tau(y_i - \mathbf{x}'_i \beta, -\mathbf{x}'_i \mathbf{w}) \mathbf{x}'_i \mathbf{w},$$

where

$$\psi_\tau(u, v) = \begin{cases} \tau - I(u < 0) & \text{if } u \neq 0 \\ \tau - I(v < 0) & \text{if } u = 0. \end{cases}$$

The optimal solution then is the value $\hat{\beta}_\tau$ where all direction-specific derivatives are positive, indicating that whenever we move from $\hat{\beta}_\tau$ towards any alternative candidate solution, the optimization criterion will increase.



Properties

One important property of the quantile regression estimate, inherited from estimated quantiles (determined from an i.i.d. sample), is its invariance under monotonic transformations. More specifically, if $\mathbf{x}'\hat{\boldsymbol{\beta}}_\tau$ is an estimate for the τ -quantile of the distribution of the response y , given covariates $\mathbf{x}$, then for any monotonically increasing transformation h the transformed estimate $h(\mathbf{x}'\hat{\boldsymbol{\beta}}_\tau)$ is an estimate for the τ -quantile of the distribution of $h(y)$. This simply follows from the fact that quantiles are invariant under monotonically increasing transformations, as the order of the data are preserved.

It is also possible to derive asymptotic results for the distribution of quantile regression coefficients under different assumptions for the error terms that may then form the basis for statistical inference. For example, in case of i.i.d. errors, the asymptotic distribution of $\hat{\boldsymbol{\beta}}_\tau$ is given by

$$\hat{\boldsymbol{\beta}}_\tau \stackrel{a}{\sim} N\left(\boldsymbol{\beta}_\tau, \frac{\tau(1-\tau)}{f_{\varepsilon_\tau}(0)^2}(\mathbf{X}'\mathbf{X})^{-1}\right).$$

Although the asymptotic covariance matrix seems to suggest that estimation of regression quantiles becomes increasingly more precise when very small ($\tau \rightarrow 0$) or large ($\tau \rightarrow 1$) quantiles are considered (due to the product $\tau(1-\tau)$ in the numerator), in fact this effect is typically dominated by the error $f_{\varepsilon_\tau}(0)$ which will usually be close to zero for such values of τ . Hence, the estimation of interior quantiles close to the median will be more precise for most of the standard distributions, while estimation of outer quantiles is associated with larger uncertainty. The asymptotic normality result can also be generalized to non i.i.d. situations (which typically fit better to situations where quantile regression is of interest). For more information, see Chap. 3 in [Koenker \(2005\)](#). While the asymptotic normal distribution provides one possibility for performing inferences on the estimated regression coefficients $\hat{\boldsymbol{\beta}}_\tau$, there are also alternatives based on the relationship between quantiles and ranks. These rank-based inferential procedures are typically preferred in practice and also allow the construction of confidence intervals by inverting rank-based tests. Again, see Chap. 3 in [Koenker \(2005\)](#).

In theory, the quantiles of the distribution of a response should be ordered such that

$$\mathbf{x}'\boldsymbol{\beta}_{\tau_1} \leq \mathbf{x}'\boldsymbol{\beta}_{\tau_2} \quad \text{for } \tau_1 \leq \tau_2$$

holds for any covariate vector $\mathbf{x}$. It can be shown that the ordering is preserved for the average covariate vector

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i,$$

when replacing the theoretical quantiles with estimated quantiles, i.e.,

$$\bar{\mathbf{x}}'\hat{\boldsymbol{\beta}}_{\tau_1} \leq \bar{\mathbf{x}}'\hat{\boldsymbol{\beta}}_{\tau_2} \quad \text{for } \tau_1 \leq \tau_2.$$

However, in general these results will not transfer to arbitrary vectors $\mathbf{x}$ and will not even hold for all observed covariate vectors. This is due to the fact that each estimate is obtained separately for each specific choice of the quantile τ . In fact, the phenomenon of crossing quantiles will always occur when the regression lines for two quantiles are not exactly parallel. Note that parallel quantile curves can only occur when the error terms are independent and identically distributed, and as such the quantile of interest merely induces a shift in the intercept parameter. While in this case no quantile-crossing will be observed, the results of quantile regressions are then only mildly interesting (since the covariate is independent of the quantile). Yet, the result for the average covariate vector is reassuring and indicates that quantile-crossing will most likely occur at the boundaries of the covariate space, where the data may be sparse.

Example 10.1 Munich Rent Index—Quantile Regression



To illustrate some simple quantile regression models, we use the data from the Munich rent index shown in Fig. 10.1. We fit three different linear models for the effect of living area and three different quadratic models for the effect of the year of construction, in each case to determine the 11 quantiles $\tau = 0.01, 0.1, 0.2, \dots, 0.8, 0.9, 0.99$ of the net rent distribution. The three different approaches are summarized as:

- Quantile regression: $y = \mathbf{x}'\boldsymbol{\beta}_\tau + \varepsilon_\tau$ as introduced in this chapter
- Homoscedastic linear model with i.i.d. Gaussian error terms: $y = \mathbf{x}'\boldsymbol{\beta} + \varepsilon, \varepsilon \sim N(0, \sigma^2)$
- Heteroscedastic linear model with independent Gaussian error terms: $y = \mathbf{x}'\boldsymbol{\beta} + \exp(\mathbf{x}'\boldsymbol{\alpha})\varepsilon, \varepsilon \sim N(0, 1)$

For the quadratic effect of the year of construction, we use an orthonormal polynomial. For both linear models, the regression quantiles are determined according to Eq. (10.2). The heteroscedastic linear model is fitted within the GAMLSS framework of [Rigby and Stasinopoulos \(2005\)](#).

The resulting estimates are shown in Fig. 10.2. For the quantile regression approach, we find a clear dependency of the specified quantile on the covariate effects, and that the quantiles correspond very well to the structure in the scatter plots. In particular, the effect of the living area almost disappears for apartments in the lower price segment (corresponding to a small value for the quantile τ), while the slope steadily increases when moving toward higher price segments (corresponding to large quantiles τ). For the year of construction, the effect is very close to linear for the lower quantiles, while it gets more and more U-shaped when moving towards higher quantiles. In this analysis, quantile-crossing does not occur, since we do not consider quantiles that are very close to each other and since the data set is relatively large (with approximately 3000 observations).

When comparing these results with those from a homoscedastic linear model, we find distinct differences. In particular, since we are assuming i.i.d. errors here, all quantile curves are exactly parallel to each other and only shifted according to the quantiles of a normal distribution. In contrast, results obtained in the heteroscedastic case are closer to the ones from the quantile regression. Note that the formula for the variance of the errors involves the exponential function and therefore induces nonlinear quantile curves, even in case of a linear model as for the living area. While the fit for living area seems to be sufficient for both the quantile regression model and the heteroscedastic linear model, the latter seems to be inadequate for describing the effect of the year of construction (at least for extreme quantiles). While large quantiles are underestimated, the curve for the lowest quantile seems to be much too low. This may be seen as an indication that, in addition to heteroscedasticity, there is also a dependence of skewness and/or kurtosis on the covariates. For these data, it appears to pay off to use a completely distribution-free approach that does not assume any specific form for the error distribution.

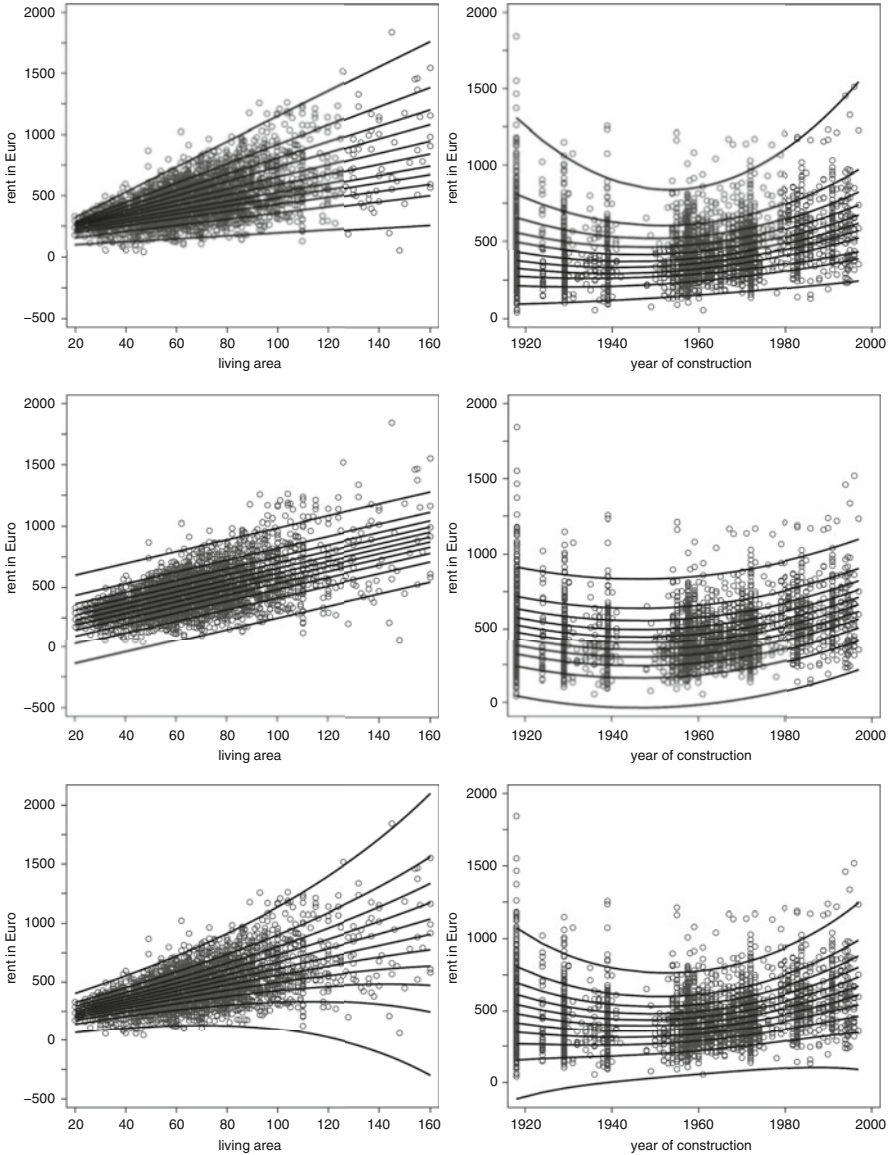


Fig. 10.2 Munich rent index: scatter plots of the rents in Euro versus living area (*left column*) and year of construction (*right panel*) together with linear/quadratic quantile regression fits for 11 quantiles (*top row*), quantiles determined from a homoscedastic linear model (*middle row*), and quantiles determined from a heteroscedastic linear model (*bottom row*)

While thus far we have only considered models consisting of one covariate with either linear or quadratic impact on the quantiles of the response, we will now consider a multiple regression model containing both effects simultaneously. This does not cause any difficulties with respect to the estimation of the regression coefficients, but the presentation

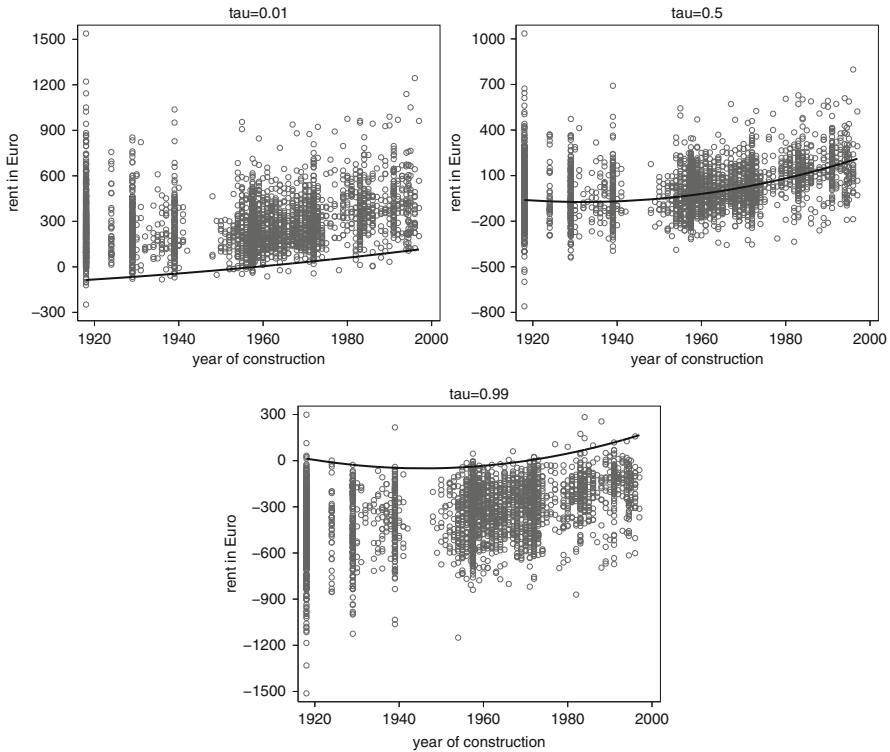


Fig. 10.3 Munich rent index: estimated effects of year of construction together with partial residuals for different quantiles

of the results is more involved. One problem is that we can no longer simply draw a scatter plot of the data along with the fits for the different quantiles, since we now must adjust for the effects of the other covariates. In the linear model, we used partial residuals to achieve this (see Sect. 3.2.3, p. 126), where all effects apart from those associated with the covariate of interest are subtracted from the response. This can also be done in quantile regression but yields different partial residuals for each value of τ . Therefore, we can no longer visualize all estimated effects simultaneously in one scatter plot. Figure 10.3 shows the estimated effect of the year of construction for three different quantiles along with the corresponding partial residuals. In all three cases, the fit appears to be quite satisfactory but with somewhat reduced curvature as compared to the results from the univariate model.

Figure 10.4 illustrates an alternative way to graphically present the results of quantile regressions, where the estimated coefficients and confidence intervals are plotted against the quantiles. The variation over the quantiles then gives indication for the necessity of quantile regression. If some or all of the estimated coefficients are almost constant, then they are independent of the quantile, and therefore only the mean but not the quantiles seem to depend on the covariate of interest. In addition, the figure shows the estimates from mean regression as a reference. Note that the magnitude of the estimated coefficients for living area and year of construction is very different since living area enters the model in untransformed linear fashion while year of construction is represented as an orthonormal polynomial.

△

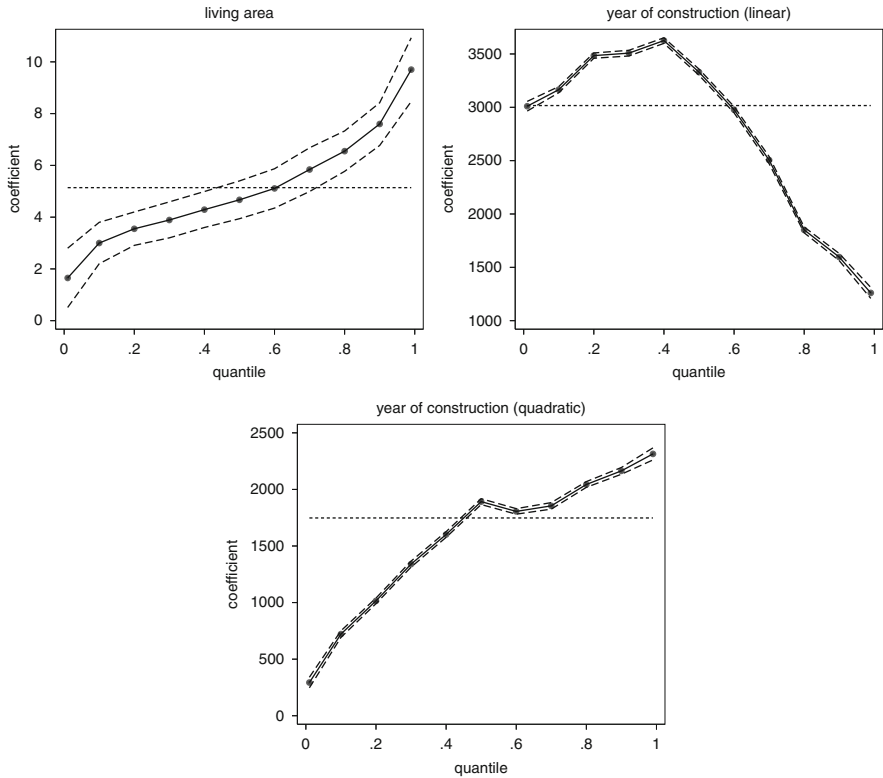


Fig. 10.4 Munich rent index: paths of estimated coefficients (*solid line*) together with 95 % confidence intervals (*dashed lines*) obtained from inverting a rank-based test for various quantiles τ . The horizontal dotted line corresponds to the least squares estimate

10.2.2 Bayesian Quantile Regression

Bayesian quantile regression has been developed utilizing the equivalence between posterior mode and maximum likelihood estimation under noninformative priors $\beta_\tau \propto \text{const}$ (Yu & Moyeed, 2001; Yue & Rue, 2011). Therefore we have to define a specific distributional assumption for the error terms (or equivalently the responses) to make the Bayesian standard machinery work. If we start with the model

$$y_i = \mathbf{x}_i' \beta_\tau + \varepsilon_{i\tau}, \quad i = 1, \dots, n,$$

we will assume independent and identically distributed errors following an asymmetric Laplace distribution, i.e., $\varepsilon_{i\tau} \mid \sigma^2$ i.i.d. $\text{ALD}(0, \sigma^2, \tau)$ with density

$$p(\varepsilon_{i\tau} \mid \sigma^2) = \frac{\tau(1-\tau)}{\sigma^2} \exp\left(-w_\tau(\varepsilon_{i\tau}, 0) \frac{|\varepsilon_{i\tau}|}{\sigma^2}\right).$$

For the responses, the error distribution induces $y_i | \boldsymbol{\beta}_\tau, \sigma^2 \sim \text{ALD}(\mathbf{x}'_i \boldsymbol{\beta}_\tau, \sigma^2, \tau)$, such that the density of the responses is given by

$$p(y_i | \boldsymbol{\beta}_\tau, \sigma^2) = \frac{\tau(1-\tau)}{\sigma^2} \exp\left(-w_\tau(y_i, \mathbf{x}'_i \boldsymbol{\beta}_\tau) \frac{|y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau|}{\sigma^2}\right).$$

It then turns out that maximizing the corresponding posterior (for fixed σ^2)

$$\begin{aligned} p(\boldsymbol{\beta}_\tau | \mathbf{y}, \sigma^2) &\propto \prod_{i=1}^n p(y_i | \boldsymbol{\beta}_\tau, \sigma^2) \\ &\propto \exp\left(-\sum_{i=1}^n w_\tau(y_i, \mathbf{x}'_i \boldsymbol{\beta}_\tau) \frac{|y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau|}{\sigma^2}\right), \end{aligned}$$

with respect to $\boldsymbol{\beta}_\tau$, is equivalent to minimizing the optimization criterion (10.5).

While the asymmetric Laplace distribution allows to conveniently express quantile regression in a Bayesian framework, it complicates inference based on Markov chain Monte Carlo simulations due to the absolute value contained in its definition. It is therefore advantageous to represent the asymmetric Laplace distribution as a scale mixture of normal distributions as suggested in [Yue and Rue \(2011\)](#): Let $z_i | \sigma^2 \sim \text{Expo}(1/\sigma^2)$, $i = 1, \dots, n$, be i.i.d. exponentially distributed with rate parameter σ^2 and

$$y_i | z_i, \boldsymbol{\beta}_\tau, \sigma^2 \sim N(\mathbf{x}'_i \boldsymbol{\beta}_\tau + \xi z_i, \sigma^2/w_i)$$

with

$$\xi = \frac{1-2\tau}{\tau(1-\tau)}, \quad w_i = \frac{1}{\delta^2 z_i}, \quad \delta^2 = \frac{2}{\tau(1-\tau)}.$$

Then the marginal distribution $y_i | \boldsymbol{\beta}_\tau, \sigma^2$ is obtained by integrating out z_i and is indeed an asymmetric Laplace distribution, i.e.,

$$y_i | \boldsymbol{\beta}_\tau, \sigma^2 \sim \text{ALD}(\mathbf{x}'_i \boldsymbol{\beta}_\tau, \sigma^2, \tau).$$

Bayesian inference can now efficiently be implemented after imputing the scale variables z_i as additional unknowns, similar to the approach that was outlined for binary regression models in Sect. 5.6.3. Basically, the resulting model is a conditionally Gaussian regression model with offsets ξz_i and weights w_i . Box 10.2 summarizes the Bayesian model and provides the full conditionals required for the Gibbs sampler. We thereby assume noninformative priors $\boldsymbol{\beta}_\tau \propto \text{const}$ and the usual inverse gamma priori for σ^2 , i.e., $\sigma^2 \sim \text{IG}(a, b)$ with hyperparameters a and b . A derivation of the full conditionals can be found at the end of this chapter on p. 618.

When comparing these results with the ones from the basic Bayesian linear model in Sect. 4.4, the only change in the full conditional for the regression coefficients is that the classical linear model is replaced by a general linear model

10.2 Bayesian Linear Quantile Regression Based on the ALD Distribution

Observation Model

The observations $y_i, i = 1, \dots, n$, are conditionally independent following an asymmetric Laplace distribution, i.e., y_i i.i.d. $\text{ALD}(\mathbf{x}'_i \boldsymbol{\beta}_\tau, \sigma^2, \tau)$. The scale mixture representation of the asymmetric Laplace distribution yields

$$y_i | z_i, \boldsymbol{\beta}_\tau, \sigma^2 \sim N(\mathbf{x}'_i \boldsymbol{\beta}_\tau + \xi z_i, \sigma^2 / w_i),$$

where

$$\xi = \frac{1 - 2\tau}{\tau(1 - \tau)}, \quad w_i = \frac{1}{\delta^2 z_i}, \quad \delta^2 = \frac{2}{\tau(1 - \tau)}.$$

Priors

$$\boldsymbol{\beta}_\tau \propto \text{const}$$

$$z_i | \sigma^2 \sim \text{Expo}(1/\sigma^2)$$

$$\sigma^2 \sim \text{IG}(a, b)$$

All priors are mutually independent.

Full Conditionals Required for the Gibbs Sampler

- Full conditional for the regression coefficients: $\boldsymbol{\beta}_\tau | \cdot \sim N(\boldsymbol{\mu}_{\boldsymbol{\beta}_\tau}, \boldsymbol{\Sigma}_{\boldsymbol{\beta}_\tau})$ with

$$\boldsymbol{\Sigma}_{\boldsymbol{\beta}_\tau} = \sigma^2 (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1}, \quad \boldsymbol{\mu}_{\boldsymbol{\beta}_\tau} = (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W} (\mathbf{y} - \xi \mathbf{z}),$$

where $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ and $\mathbf{z} = (z_1, \dots, z_n)'$.

- Full conditional for the scale parameters:

$$z_i^{-1} | \cdot \sim \text{InvGauss} \left(\sqrt{\frac{\xi^2 + 2\delta^2}{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)^2}}, \frac{\xi^2 + 2\delta^2}{\sigma^2 \delta^2} \right),$$

refer to Definition B.7 in Appendix B.1 for the inverse Gaussian distribution.

- Full conditional for the error variance:

$$\sigma^2 | \cdot \sim \text{IG} \left(a + \frac{3n}{2}, b + \frac{1}{2} \sum_{i=1}^n w_i (y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau - \xi z_i)^2 + \sum_{i=1}^n z_i \right).$$

Software

Software package BayesX (see also the R interface R2BayesX)

with weights w_i and offset ξz_i . This also indicates that it is easy to extend the Bayesian quantile regression model beyond linear regression specifications, as outlined in the following section by including, for example, Bayesian P-splines or spatial effects. The full conditional for the error variance also changes since the exponential prior of the weights depends on σ^2 .

The presented Bayesian quantile regression approach relies on the asymmetric Laplace distribution as an auxiliary error distribution that yields a formal equivalence between posterior modes and usual quantile regression estimates. However, in a strict sense, this auxiliary error distribution can never be simultaneously true for all quantiles since only one quantile of the error distribution can in fact be zero. Consequently, the resulting point estimates are often close to those from frequentist analyses, while the corresponding credible intervals may be misleading when the error distribution is (greatly) misspecified. However, the Bayesian quantile regression approach still has considerable advantages with respect to generalized model specifications as will be discussed in the next section. Other Bayesian quantile regression approaches try to circumvent the auxiliary error distribution problem by including the error distribution as a part of the estimation problem and to determine quantiles from the estimated error distribution. See, for example, [Reich, Bondell, and Wang \(2010\)](#) and [Kottas and Krnjajic \(2009\)](#) for approaches based on mixtures of Gaussians and Dirichlet process mixtures, respectively.

10.3 Additive Quantile Regression

Additive quantile regression models result when replacing the assumed linear effects of covariates thus far, with potentially nonlinear effects, similar to those found in Chaps. 8 and 9. An approach for estimating nonlinear quantile functions $f_\tau(z_i)$ of continuous covariates z_i in the scatter plot smoothing model

$$y_i = f_\tau(z_i) + \varepsilon_{i\tau}$$

that still fits into the framework of linear programming (which is usually employed for estimating linear quantile regression models) has been suggested by [Koenker, Ng, and Portnoy \(1994\)](#). Such an approach relies on the fitting criterion

$$\arg \min_{f_\tau} \sum_{i=1}^n w_\tau(y_i, f_\tau(z_i)) |y_i - f_\tau(z_i)| + \lambda V(f'_\tau). \quad (10.7)$$

This criterion is inspired by the penalized least squares criterion employed in the derivation of smoothing splines (Sect. 8.1.4, p. 448) but relies on asymmetrically weighted absolute errors instead of least squares. Additionally it considers a somewhat different penalty that fits well with the absolute value criterion. More specifically, $V(f'_\tau)$ denotes the total variation of the derivative f'_τ defined as

$$V(f'_\tau) = \sup \sum_{i=1}^n |f'_\tau(z_{i+1}) - f'_\tau(z_i)|,$$

where the sup is taken over all partitions $a \leq z_1 < \dots < z_n < b$. For twice continuously differentiable functions f_τ , the total variation penalty can be written as

$$V(f'_\tau) = \int |f''_\tau(z)| dz$$

indicating that the penalty results from replacing the squared second derivative (applied in the context of smoothing splines) with the absolute second derivative. Utilizing the squared second derivative would inhibit the use of linear programming techniques since it would lead to a combination of an L_1 norm for the error terms and an L_2 norm for the penalty. In contrast, [Koenker et al. \(1994\)](#) showed that the solution to Eq. (10.7) can still be obtained by linear programming when considering a somewhat larger function space that also comprises functions with derivatives existing almost everywhere. Within this function space, the minimizer of Eq. (10.7) is a piecewise linear spline function with knots at the observations z_i , similar to those found with the natural cubic spline resulting in the context of smoothing splines.

This approach can also be extended to additive models that consist of more than one nonparametric function or the estimation of surfaces based on triograms [Koenker and Mizera \(2004\)](#). However, it is typically difficult to determine the smoothing parameters along with the estimated functions in an automatic and data-driven way. As a consequence, alternative approaches based on boosting or the Bayesian approach to quantile regression have been suggested, for example, in [Fenske et al. \(2011\)](#), [Yue and Rue \(2011\)](#), or [Waldmann, Kneib, Lang, and Yue \(2012\)](#).

For the boosting approach, we basically have to combine the penalized least squares base-learners that were introduced for structured additive regression in Sect. 9.7, with the loss function defining quantile regression estimates. Hence, the only change relates to different negative gradients employed in the boosting algorithm. More specifically, in the algorithm summarized in Box 9.6, the negative gradients are replaced by Eq. (10.6) where η can now be any structured additive predictor. All the remaining parts of the algorithm can be reused without any modifications.

For Bayesian additive quantile regression, we can extend the Gibbs-sampler outlined in the previous section in a similar spirit as in Sect. 9.6.3 for structured additive regression, relying on Markov chain Monte Carlo simulation techniques. Most importantly, the full conditionals for nonparametric effects represented as $V_j \gamma_j$ are now given by

$$\gamma_j | \cdot \sim N(\mathbf{m}_j, \boldsymbol{\Sigma}_j),$$

with expectation and covariance matrix

$$\begin{aligned} \mathbf{m}_j &= E(\gamma_j | \cdot) = \left(\mathbf{V}'_j \mathbf{W} \mathbf{V}_j + \frac{\sigma^2}{\tau_j^2} \mathbf{K}_j \right)^{-1} \mathbf{V}_j \mathbf{W}' (\mathbf{y} - \boldsymbol{\eta}_{-j}^{struct} - \xi \mathbf{z}) \\ \boldsymbol{\Sigma}_j &= \text{Cov}(\gamma_j | \cdot) = \sigma^2 \left(\mathbf{V}'_j \mathbf{W} \mathbf{V}_j + \frac{\sigma^2}{\tau_j^2} \mathbf{K}_j \right)^{-1}, \end{aligned}$$

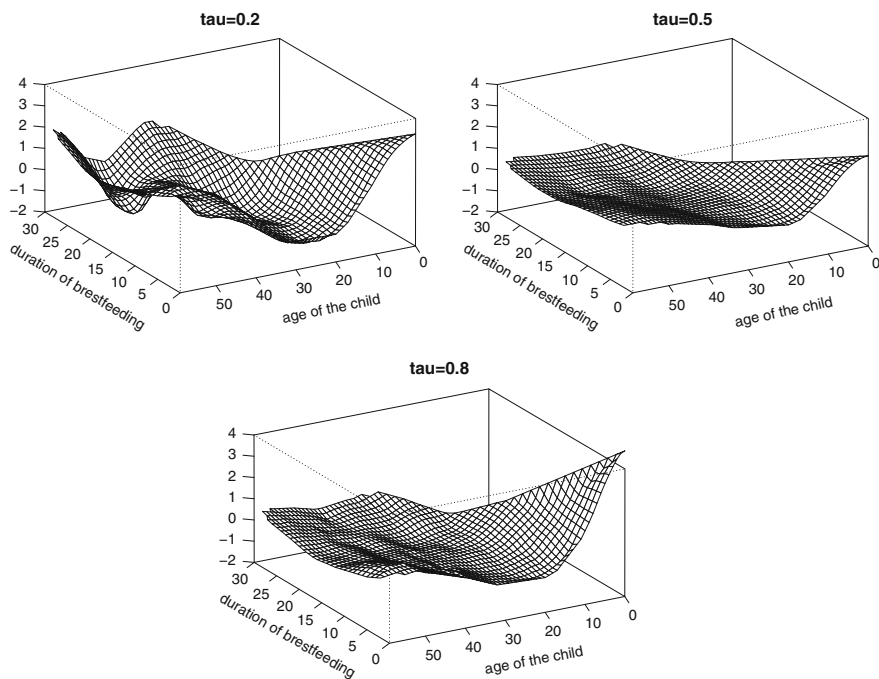


Fig. 10.5 Malnutrition in Zambia: interaction of the child's age and duration of breast-feeding obtained in a geoadditive Bayesian quantile regression model for $\tau = 0.2$ (top left), $\tau = 0.5$ (top right), and $\tau = 0.8$ (bottom)

where $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ and $\mathbf{z} = (z_1, \dots, z_n)'$ as in Box 10.2. Similarly, the full conditional for the error variance has to be adjusted while the full conditionals for the smoothing variances remain unchanged.

Example 10.2 Childhood Malnutrition in Zambia—Geoadditive Bayesian Quantile Regression



We illustrate the results acquired from a Bayesian geoadditive quantile regression model based on the asymmetric Laplace distribution as auxiliary error distribution by extending the case study presented in Sect. 9.8. Therefore, we estimate a quantile regression model

$$\begin{aligned} zscore_i = & f_{1\tau}(m_agebirth_i) + f_{2\tau}(m_height_i) + f_{3\tau}(m_bmi_i) \\ & + f_{4\tau}(c_breastf_i, c_age_i) + f_{\text{spatr}}(s_i) + \mathbf{x}'_i \boldsymbol{\beta}_\tau + \varepsilon_{i\tau} \end{aligned}$$

comprising a kriging term $f_{4\tau}(c_breastf, c_age)$ for the interaction of duration of breast-feeding and age of the child, penalized splines $f_{1\tau}(m_agebirth)$, $f_{2\tau}(m_height)$, and $f_{3\tau}(m_bmi)$ for the nonlinear effects of the body mass index, height, and age of the mother, a Markov random field $f_{\text{spatr}}(s)$ for the spatial effects of the districts s , and parametric effects $\mathbf{x}'_i \boldsymbol{\beta}_\tau$ for the remaining (categorical) covariates. All hyperparameter settings (such as number of knots and degrees of the spline) have been chosen analogously to those in the case study in the previous chapter.

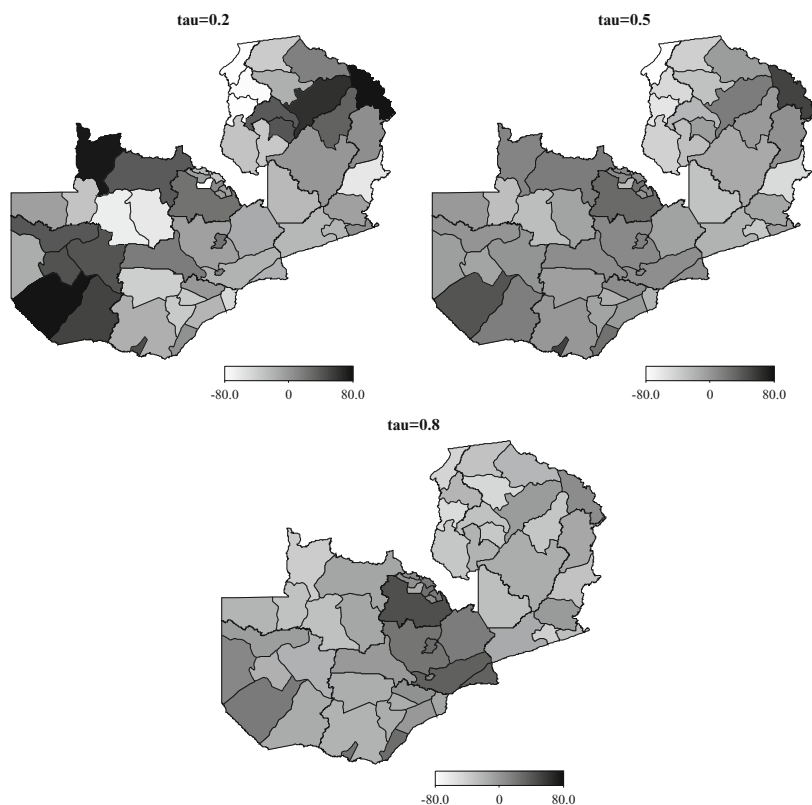


Fig. 10.6 Malnutrition in Zambia: spatial effect obtained in a geoaddditive Bayesian quantile regression model for $\tau = 0.2$ (top left), $\tau = 0.5$ (top right), and $\tau = 0.8$ (bottom)

To implement the Bayesian approach, we make use of the latent Gaussian representation for the asymmetric Laplace distribution as introduced in Sect. 10.2.2. Refer to [Waldmann et al. \(2012\)](#) for details. In the following, we discuss only some exemplary results for the 20 %, 50 %, and 80 % quantiles.

Figure 10.5 shows estimates for the interaction effect between the age of the child and the duration of breast-feeding. While the principal form of the effect is not too different between the quantiles, and it also matches closely with the results from the mean regression model presented in the last chapter, the magnitude of the effects is much larger for the 20 % and the 80 % quantile as compared to the median regression results. Similarly, the 20 % quantile estimates for the spatial effect (Fig. 10.6) closely resemble the form of the spatial effect for the median but again show a larger magnitude. In contrast, the 80 % quantile also shows a different spatial pattern, indicating that the spatial distribution in case of well-nourished children is different from the one for malnourished children and children with average nutritional status. Finally, Fig. 10.7 shows estimates for the effect of the body mass index of the mother. There seems to be a slight indication for a nonlinear effect but with some uncertainty, especially for large and small values of the body mass index. Note, however, that the estimated credible intervals have to be treated with caution since they rely

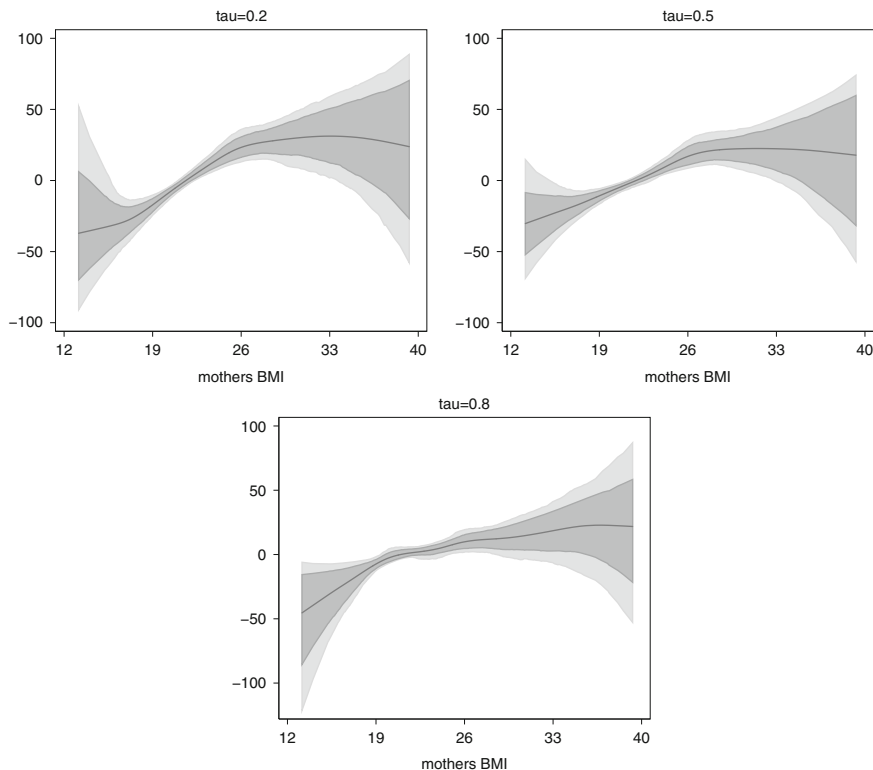


Fig. 10.7 Malnutrition in Zambia: nonparametric effect of the body mass index of the mother obtained in a geoadditive Bayesian quantile regression model for $\tau = 0.2$ (top left), $\tau = 0.5$ (top right), and $\tau = 0.8$ (bottom)

on the assumption of an asymmetric Laplace distribution for the errors which will usually not be fulfilled in practice.

△

10.4 Bibliographic Notes and Proofs

10.4.1 Bibliographic Notes

Non-crossing Quantiles

Approaches that circumvent the problem of crossing quantiles are typically based on joint estimation for several quantiles. The simplest example is the location-scale model

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + \exp(\mathbf{z}_i' \boldsymbol{\alpha}) \varepsilon_i, \quad \text{Var}(\varepsilon_i) = 1,$$

where not only the mean but also the standard deviation of the response variable depend on covariates. This model basically reflects the situation of heteroscedasticity, but is less suited when more general effects of the covariates, e.g., on the skewness or kurtosis, are present as well. Estimation of the location-scale model can either be based on a distributional assumption for ε_i (such as a normal distribution) or include an additional estimation step for fitting the quantile curves to the data as suggested in different variants in [He \(1997\)](#) and [Schnabel and Eilers \(2012\)](#). A more sophisticated approach is quantile regression sheets where the regression coefficient β_τ corresponding to a specific covariate is treated as a function $\beta(\tau)$ that is estimated as a penalized spline with penalization along the quantile-domain; see [Schnabel and Eilers \(2012\)](#).

Expectile Regression

Some of the difficulties associated with extending linear quantile regression to more complex settings and developing completely data-driven estimation approaches for these settings can be attributed to the fact that the absolute deviation criterion (10.5) is not differentiable and therefore requires more specialized optimization approaches, such as linear programming or boosting. As a computationally attractive alternative, [Newey and Powell \(1987\)](#) have proposed expectiles that replace asymmetrically weighted absolute deviations with asymmetrically weighted squared deviations yielding the optimality criterion

$$\hat{\beta}_\tau = \arg \min_{\beta_\tau} \sum_{i=1}^n w_\tau(y_i, \eta_{i\tau})(y_i - \eta_{i\tau})^2.$$

This approach has recently gained considerable interest since the solutions can be computed by simple iteratively weighted least squares updates

$$\hat{\beta}_\tau^{(t+1)} = (X'W_\tau^{(t)}X)^{-1}X'W_\tau^{(t)}y,$$

with iteratively recomputed weights

$$W_\tau^{(t)} = \text{diag}(w_\tau(y_1, \eta_{1\tau})^{(t)}, \dots, w_\tau(y_n, \eta_{n\tau})^{(t)}).$$

This has particular advantages when considering complex models defined, for example, in terms of several nonlinear functions and quadratic penalties, as in the context of structured additive regression. In particular, the smoothing parameters can then still be automatically computed, for example, utilizing the mixed model representation of structured additive regression models; see [Sobotka and Kneib \(2012\)](#) for details. Although expectiles do not enjoy the intuitive interpretation of quantiles, they still provide a possibility to characterize the conditional distribution of a response distribution and therefore contain similar information as a set of quantile regressions. An implementation of structured additive expectile regression models is provided in the R package `expectreg`.

10.4.2 Proofs

Optimality Criterion for Quantiles (p. 599)

We show the equivalence of the implicit definition of quantiles and the optimality criterion (10.5) for a continuous random variable y with strictly increasing cumulative distribution function $F(y)$ and density $f(y)$ such that the τ -quantile q_τ is unique. Therefore, we first rewrite the expectation of the minimization criterion in terms of the density $f(y)$ as

$$\begin{aligned} E(w_\tau(y)|y - q|) &= \int_{-\infty}^{\infty} w_\tau(y)|y - q|f(y)dy \\ &= \int_{-\infty}^q (1 - \tau)(y - q)f(y)dy + \int_q^{\infty} \tau(y - q)f(y)dy. \end{aligned}$$

The minimum can then be found by differentiating with respect to q and setting the derivative to zero. As the limits of integration depend on q , we have to apply Leibniz integral rule, yielding

$$\begin{aligned} \frac{\partial}{\partial q} E(w_\tau(y)|y - q|) &= -(1 - \tau) \int_{-\infty}^{q_\tau} f(y)dy + \tau \int_{q_\tau}^{\infty} f(y)dy \\ &= - \int_{-\infty}^{q_\tau} f(y)dy + \tau \int_{-\infty}^{\infty} f(y)dy \\ &= - \int_{-\infty}^{q_\tau} f(y)dy + \tau. \end{aligned}$$

Setting this expression equal to zero, we obtain

$$\tau = \int_{-\infty}^{q_\tau} f(y)dy = F(q_\tau) = P(y \leq \tau).$$

Since we assumed that the τ -quantile is unique, this also implies $P(y \geq q_\tau) = 1 - \tau$ and therefore proves the equivalence between both definitions.

Derivation of the Full Conditionals in the Bayesian Linear Quantile Regression Model of Box 10.2 on p. 611

We derive the full conditionals for the model

$$y | z, \beta_\tau, \sigma^2 \sim N(X\beta_\tau + \xi z, \sigma^2 W^{-1}), \quad W = \text{diag}(w_1, \dots, w_n), \quad z = (z_1, \dots, z_n)'$$

with

$$\xi = \frac{1 - 2\tau}{\tau(1 - \tau)}, \quad w_i = \frac{1}{\delta^2 z_i}, \quad \delta^2 = \frac{2}{\tau(1 - \tau)},$$

and the priors

$$\boldsymbol{\beta}_\tau \propto \text{const}, \quad z_i | \sigma^2 \sim \text{Expo}(1/\sigma^2), \quad \sigma^2 \sim \text{IG}(a, b).$$

Using $\boldsymbol{\Sigma}_{\boldsymbol{\beta}_\tau} = \sigma^2(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}$ and $\boldsymbol{\mu}_{\boldsymbol{\beta}_\tau} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}(\mathbf{y} - \xi\mathbf{z})$, we obtain

$$\begin{aligned} p(\boldsymbol{\beta}_\tau | \cdot) &\propto p(\mathbf{y} | \mathbf{z}, \boldsymbol{\beta}_\tau, \sigma^2) \\ &\propto \exp\left\{-\frac{1}{2\sigma^2}(\mathbf{X}\boldsymbol{\beta}_\tau + \xi\mathbf{z} - \mathbf{y})'\mathbf{W}(\mathbf{X}\boldsymbol{\beta}_\tau + \xi\mathbf{z} - \mathbf{y})\right\} \\ &\propto \exp\left\{-\frac{1}{2}\boldsymbol{\beta}_\tau' \frac{1}{\sigma^2} \mathbf{X}'\mathbf{W}\mathbf{X}\boldsymbol{\beta}_\tau + \frac{1}{\sigma^2}\boldsymbol{\beta}_\tau' \mathbf{X}'\mathbf{W}(\mathbf{y} - \xi\mathbf{z})\right\} \\ &= \exp\left\{-\frac{1}{2}\boldsymbol{\beta}_\tau' \boldsymbol{\Sigma}_{\boldsymbol{\beta}_\tau}^{-1}\boldsymbol{\beta}_\tau + \boldsymbol{\beta}_\tau' \boldsymbol{\Sigma}_{\boldsymbol{\beta}_\tau}^{-1}(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}(\mathbf{y} - \xi\mathbf{z})\right\} \\ &= \exp\left\{-\frac{1}{2}\boldsymbol{\beta}_\tau' \boldsymbol{\Sigma}_{\boldsymbol{\beta}_\tau}^{-1}\boldsymbol{\beta}_\tau + \boldsymbol{\beta}_\tau' \boldsymbol{\Sigma}_{\boldsymbol{\beta}_\tau}^{-1}\boldsymbol{\mu}_{\boldsymbol{\beta}_\tau}\right\} \end{aligned}$$

for the full conditional of $\boldsymbol{\beta}_\tau$. According to Definition B.20 in Appendix B.3.1 this is a multivariate normal distribution with mean $\boldsymbol{\mu}_{\boldsymbol{\beta}_\tau}$ and covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\beta}_\tau}$.

The full conditional for σ^2 is given by

$$\begin{aligned} p(\sigma^2 | \cdot) &\propto p(\mathbf{y} | \mathbf{z}, \boldsymbol{\beta}_\tau, \sigma^2) p(\mathbf{z} | \sigma^2) p(\sigma^2) \\ &\propto \left(\frac{1}{\sigma^2}\right)^{\frac{n}{2}} \exp\left\{-\frac{1}{\sigma^2} \frac{1}{2}(\mathbf{X}\boldsymbol{\beta}_\tau + \xi\mathbf{z} - \mathbf{y})'\mathbf{W}(\mathbf{X}\boldsymbol{\beta}_\tau + \xi\mathbf{z} - \mathbf{y})\right\} \\ &\quad \prod_{i=1}^n \frac{1}{\sigma^2} \exp\left(-\frac{1}{\sigma^2} z_i\right) \\ &\quad \left(\frac{1}{\sigma^2}\right)^{a-1} \exp\left(-\frac{1}{\sigma^2} b\right) \\ &= \left(\frac{1}{\sigma^2}\right)^{n+\frac{n}{2}+a-1} \exp\left\{-\frac{1}{\sigma^2}\left(b + \frac{1}{2}(\mathbf{X}\boldsymbol{\beta}_\tau + \xi\mathbf{z} - \mathbf{y})'\mathbf{W}(\mathbf{X}\boldsymbol{\beta}_\tau + \xi\mathbf{z} - \mathbf{y}) + \sum_{i=1}^n z_i\right)\right\}. \end{aligned}$$

This is the density of an inverse gamma distribution with parameters

$$a' = a + \frac{3n}{2} \text{ and } b' = b + \frac{1}{2}(\mathbf{X}\boldsymbol{\beta}_\tau + \xi\mathbf{z} - \mathbf{y})'\mathbf{W}(\mathbf{X}\boldsymbol{\beta}_\tau + \xi\mathbf{z} - \mathbf{y}).$$

Finally the full conditionals for the latent variables z_i are obtained as

$$\begin{aligned}
p(z_i | \cdot) &\propto p(y_i | z_i, \boldsymbol{\beta}_\tau, \sigma^2) p(z_i | \sigma^2) \\
&\propto \frac{1}{\sqrt{z_i}} \exp \left\{ -\frac{1}{2} \frac{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau - \xi z_i)^2}{\delta^2 z_i \sigma^2} \right\} \exp \left(-\frac{1}{\sigma^2} z_i \right) \\
&= \frac{1}{\sqrt{z_i}} \exp \left\{ -\frac{1}{2} \frac{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)^2 - 2(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau) \xi z_i + z_i^2 (\xi^2 + 2\delta^2)}{\delta^2 z_i \sigma^2} \right\} \\
&= \frac{1}{\sqrt{z_i}} \exp \left\{ -\frac{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)^2}{2\sigma^2 \delta^2} \cdot \frac{1 - \frac{2\xi z_i}{y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau} + z_i^2 \frac{\xi^2 + 2\delta^2}{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)^2}}{z_i} \right\} \\
&= \frac{1}{\sqrt{z_i}} \exp \left\{ -\frac{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)^2}{2\sigma^2 \delta^2} \cdot \frac{(z_i^{-1})^2 - \frac{2\xi}{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)} z_i^{-1} + \mu_i^2}{z_i^{-1}} \right\} \\
&= \frac{1}{\sqrt{z_i}} \exp \left\{ -\frac{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)^2 (\xi^2 + 2\delta^2)}{2\sigma^2 \delta^2 (\xi^2 + 2\delta^2)} \cdot \frac{(z_i^{-1})^2 - \frac{2\xi}{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)} z_i^{-1} + \mu_i^2}{z_i^{-1}} \right\} \\
&= \frac{1}{\sqrt{z_i}} \exp \left\{ -\frac{\lambda}{2\mu_i^2} z_i \cdot \left(\frac{1}{z_i^2} - \frac{2\xi}{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)} \frac{1}{z_i} + \mu_i^2 \right) \right\} \\
&\propto \frac{1}{\sqrt{z_i}} \exp \left\{ -\frac{\lambda}{2\mu_i^2} z_i \cdot \left(\frac{1}{z_i^2} - 2\mu_i \frac{1}{z_i} + \mu_i^2 \right) \right\} \\
&= \frac{1}{\sqrt{z_i}} \exp \left\{ -\frac{\lambda}{2\mu_i^2} z_i \cdot \left(\frac{1}{z_i} - \mu_i \right)^2 \right\}
\end{aligned}$$

where

$$\mu_i = \left(\frac{\xi^2 + 2\delta^2}{(y_i - \mathbf{x}'_i \boldsymbol{\beta}_\tau)^2} \right)^{1/2} \quad \text{and} \quad \lambda = \frac{\xi^2 + 2\delta^2}{\sigma^2 \delta^2}.$$

This is not a standard distribution. However, it turns out that the distribution of $1/z_i$ is inverse Gaussian with location parameter μ_i and scale parameter λ . Updating of z_i is then obtained by first sampling $1/z_i$ from the inverse Gaussian distribution and then inverting the result to obtain z_i .

To derive the distribution of $1/z_i$ we apply the change of variables Theorem B.1 of Appendix B.1. With $g(z_i) = 1/z_i$, $g^{-1}(z_i) = z_i$ and $g'(z_i) = -1/z_i^2$, we obtain

$$p(1/z_i | \cdot) \propto \left(\frac{1}{(1/z_i)^3} \right)^{1/2} \exp \left(-\frac{\lambda}{2\mu_i (1/z_i)} (1/z_i - \mu_i)^2 \right)$$

which has the form of the proposed inverse Gaussian distribution.